Class	Index Number	Name
25S		

ST. ANDREW'S JUNIOR COLLEGE
JC 2 2026
Preliminary Examination

PHYSICS, Higher 2 **9478/03**

Paper 3 Longer Structured Questions

2 hours

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Section A: answer **all** questions.
- Section B: answer **one** question.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, index number and class in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- You may use an approved calculator.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

For Examiner's Use	
Section A	
1	**/ 6**
2	**/ 5**
3	**/ 8**
4	**/ 10**
5	**/ 6**
6	**/ 9**
7	**/ 11**
Section B	
8	**/ 20**
9	**/ 20**
Total	**/ 75**

This document has **27** pages.

Data

speed of light in free space	$c = 3.00 \times 10^{8}$ m s^{-1}
permeability of free space	$\mu_o = 4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	$\varepsilon_o = 8.85 \times 10^{-12}$ F m^{-1}
	$(\frac{1}{4\pi\varepsilon_o} = 8.99 \times 10^{9}$ m F$^{-1})$
elementary charge	$e = 1.60 \times 10^{-19}$ C
Planck constant	$h = 6.63 \times 10^{-34}$ J s
unified atomic mass constant	$u = 1.66 \times 10^{-27}$ kg
rest mass of electron	$m_e = 9.11 \times 10^{-31}$ kg
rest mass of proton	$m_p = 1.67 \times 10^{-27}$ kg
molar gas constant	$R = 8.31$ J K^{-1} mol^{-1}
Avogadro constant	$N_A = 6.02 \times 10^{23}$ mol^{-1}
Boltzmann constant	$k = 1.38 \times 10^{-23}$ J K^{-1}
gravitational constant	$G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}
acceleration of free fall	$g = 9.81$ m s^{-2}

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$
	$v^2 = u^2 + 2as$
work done on / by a gas	$W = p\,\Delta V$
pressure	$p = \frac{F}{A}$
gravitational potential	$\phi = \frac{-Gm}{r}$
temperature	$T/\text{K} = T/{}^{\circ}\text{C} + 273.15$
pressure of an ideal gas	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$

mean translational kinetic energy of an ideal gas particle	$E = \frac{3}{2}kT$
displacement of particle in s.h.m.	$x = x_o \sin \omega t$
velocity of particle in s.h.m.	$v = v_0 \cos \omega t = \pm \omega \sqrt{x_0^2 - x^2}$
electric current	$I = nAvq$
resistors in series	$R = R_1 + R_2 + ...$
resistors in parallel	$1 / R = 1 / R_1 + 1 / R_2 + ...$
capacitors in series	$1 / C = 1 / C_1 + 1 / C_2 + ...$
capacitors in parallel	$C = C_1 + C_2 + ...$
energy in a capacitor	$U = \frac{1}{2} QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2} C V^2$
charging a capacitor	$Q = Q_o[1 - e^{-\frac{t}{\tau}}]$
discharging a capacitor	$Q = Q_o e^{-\frac{t}{\tau}}$
RC time constant	$\tau = RC$
electric potential	$V = \frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	$x = x_o \sin \omega t$
magnetic flux density due to a long straight wire	$B = \frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	$B = \frac{\mu_0 N I}{2r}$
magnetic flux density due to a long solenoid	$B = \mu_0 n I$
energy states for quantum particle in a box	$E_n = \frac{h^2}{8mL^2} n^2$
radioactive decay	$x = x_o e^{-\lambda t}$
radioactive decay constant	$\lambda = \frac{\ln 2}{t_{1/2}}$

Section A

Answer **all** the questions in this section in the spaces provided

1 A rigid uniform beam of weight *W* is connected to a fixed support by a hinge as shown in Fig. 1.1.

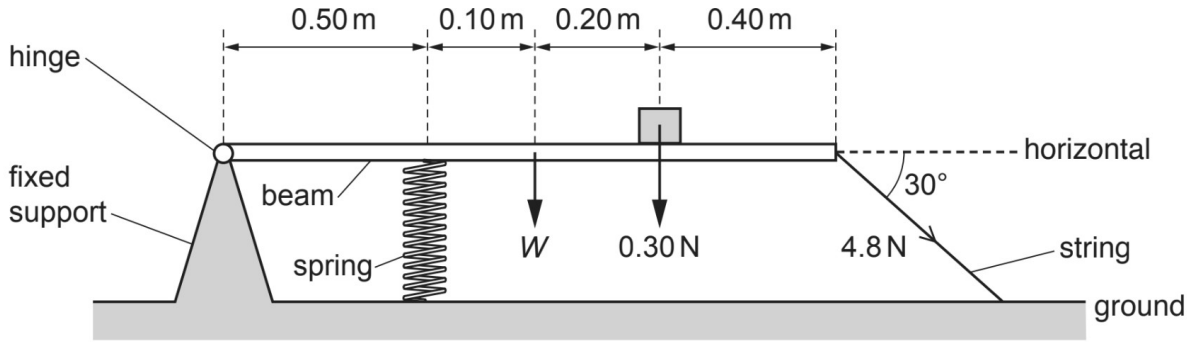


Fig. 1.1 (not to scale)

A compressed spring exerts a vertical force of 8.2 N on the horizontal beam. A block of weight 0.30 N rests on the beam. The right end of the beam is connected to the ground by a string at an angle of 30° to the horizontal. The tension in the string is 4.8 N. The distances along the beam are shown in Fig. 1.1.

The beam is in equilibrium. Assume that the hinge is frictionless.

(a) **(i)** Determine the weight *W* of the beam.

W = ... N [3]

(ii) Calculate the horizontal force exerted on the beam by the hinge.

force = ... N [1]

(b) The compressed spring obeys Hooke's law and stores an elastic potential energy of 0.32 J.

Calculate the compression of the spring.

compression = ...m [2]

[Total: 6]

2 Trolley A moves at constant velocity towards trolley B, which is initially moving at a lower constant speed in the same direction.

Fig. 2.1 shows the trolleys at time $t = 0$.



Fig. 2.1

Table 2.1 shows data for the trolleys.

Table 2.1

trolley	mass / kg	initial speed / m s^{-1}
A	0.25	0.48
B	0.75	0.12

The two trolleys collide elastically and then separate. Resistive forces are negligible.

Fig. 2.2 shows the variation with time t of the velocity v for trolley B.

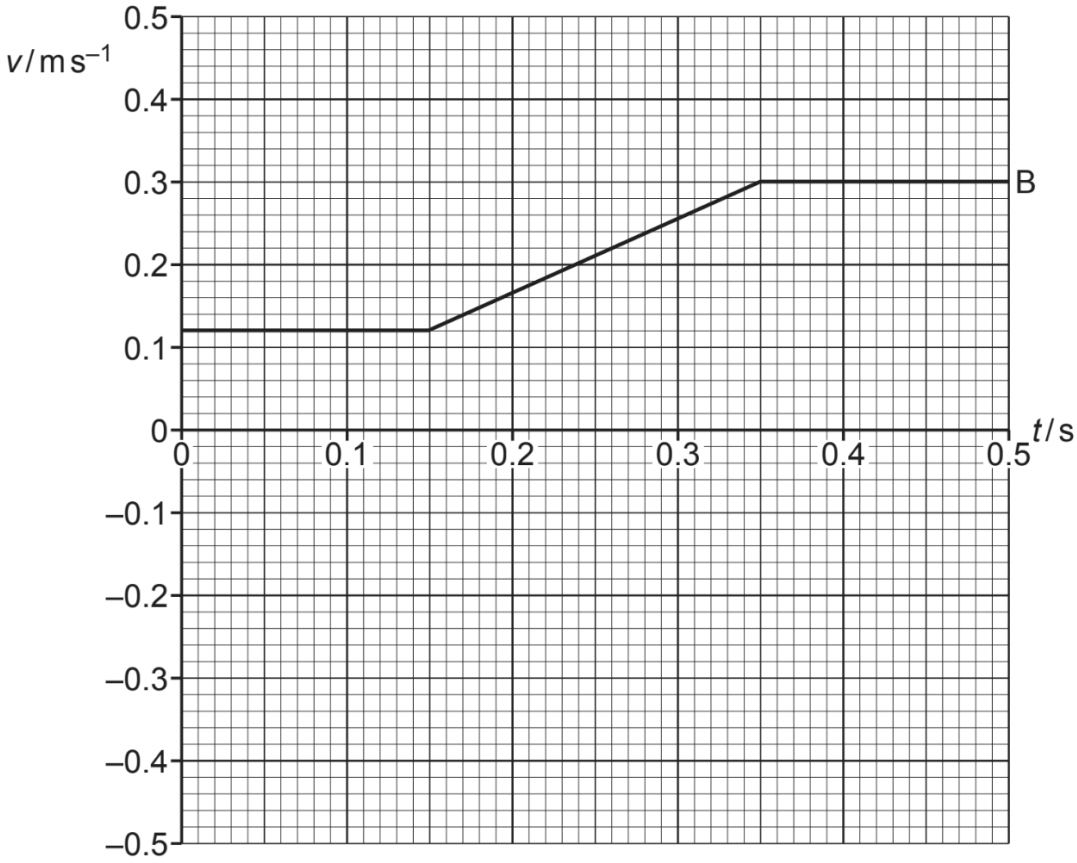


Fig. 2.2

Determine

(a) the acceleration of trolley B during the collision

acceleration of B = ... m s^{-2} [2]

(b) the magnitude and direction of the final velocity of trolley A.

magnitude = ... m s^{-1}

direction ..... [3]

[Total: 5]

3 **(a)** A satellite of mass m orbits a planet of mass M in a circular orbit, with orbital radius r.

Show that its kinetic energy E_K and gravitational potential energy E_P are related by

$$E_K = -\frac{E_P}{2}.$$

[2]

(b) The variation of gravitational potential ϕ with distance r from the centre of Jupiter is shown in Fig. 3.1.

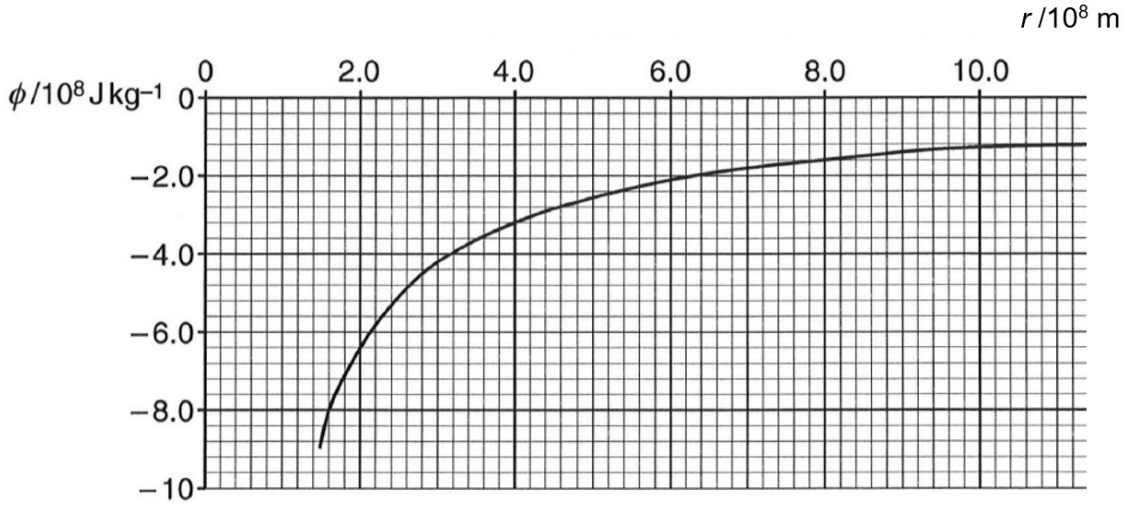


Fig. 3.1

A satellite orbiting Jupiter has a mass of 8.93×10^{22} kg and orbital radius 4.0×10^8 m.

Determine the total energy of the satellite.

total energy = …… J [3]

(c) Two identical isolated uniform spheres X and Y each have radius *R*. The centres of the

spheres are separated by distance L, as shown in Fig. 3.2.

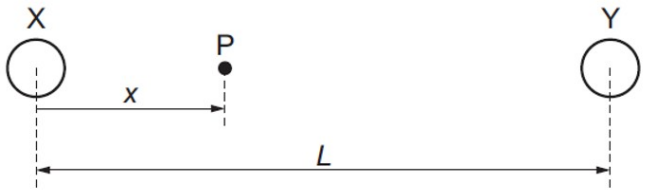


Fig. 3.2

Point P lies on the line joining the centres of X and Y, and is at a variable displacement x from the centre of sphere X.

The gravitational field strength at the surface of each sphere is g_o.

On Fig. 3.3, sketch the variation with x of the gravitational field strength g at point P between $x = R$ and $x = L - R$.

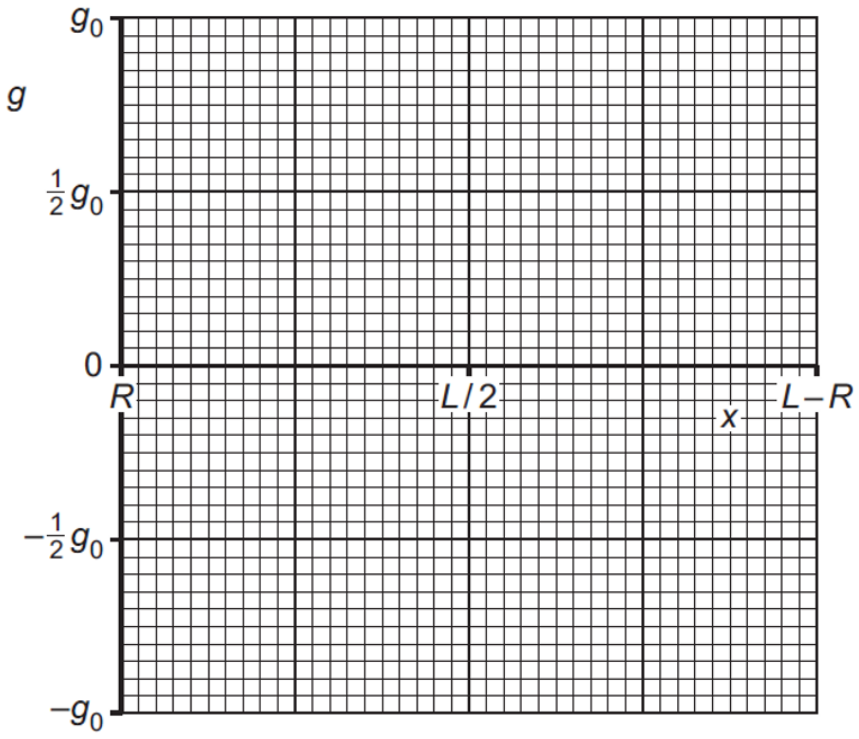


Fig. 3.3

[3]

[Total: 8]

4 **(a)** **(i)** With reference to the vibrations involved, state the difference between *transverse* and *longitudinal* waves.

.....
.....

.....
.....

..... [2]

(ii) State an example of each.

example of a transverse wave: .....

example of a longitudinal wave: ..... [1]

(b) **(i)** State what is meant by the diffraction of a wave.

……..

..... [1]

(ii) Explain how diffraction is essential for observable double-slit interference to occur. Sketch a diagram to help your answer to be clearer.

……..

……..

..... [2]

(c) Certain types of electromagnetic waves are used in radar systems where they are passed along tubes called waveguides. One part of the system can be a double tube as shown below, where the lengths of the wave paths in the tube are 18 cm and 31 cm as shown.

The wavelength of the radar waves in the waveguide is 4.0 cm.

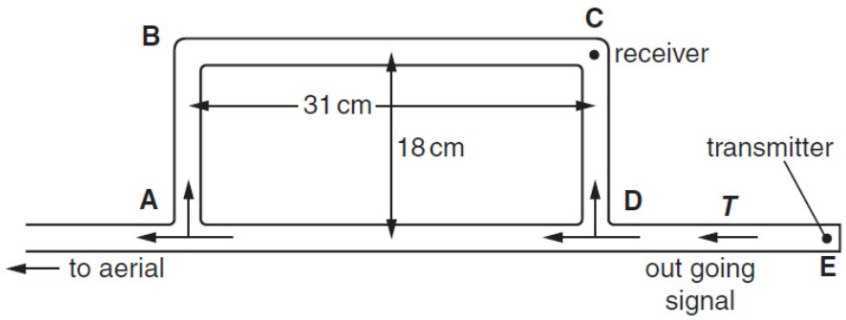


Fig. 4.1

(i) A wave ***T*** travelling from a transmitter at **E** in the direction shown in Fig. 4.1. It reaches a receiver at **C** by paths **DABC** and **DC**.

Calculate the path length of each of these paths.

path length **DC** = ... cm; path length **DABC** ..... cm [1]

(ii) Determine the phase difference between these two waves arriving at **C** and comment on the amplitude of the resultant wave at **C**.

phase difference = …… [2]

..... [1]

[Total: 10]

5 **(a)** A beam of vertically polarised light of wavelength 540 nm is incident normally on a diffraction grating, as shown in Fig. 5.1.

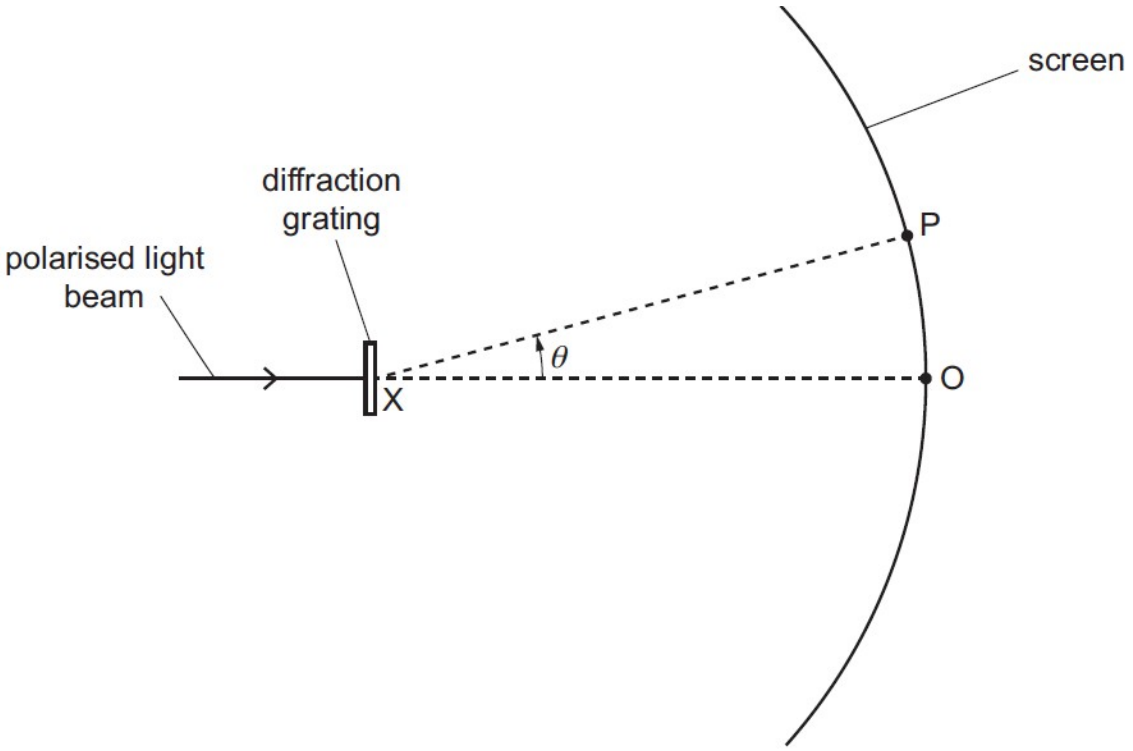


Fig. 5.1 (not to scale)

The diffraction grating has a line spacing of 5.0×10^{-6} m.

The light transmitted by the diffraction grating illuminates a circular screen. The diffraction grating is at the centre X of the circle. The central bright fringe is formed at point O on the screen and has intensity I_0.

P is a point on the screen where the line XP is at a variable angle θ to the line XO. The intensity I of light on the screen at P varies with θ.

(i) Show that the angle θ at which the first-order bright fringe is formed is 6.2°. [1]

(ii) Determine the value of θ at which the second-order bright fringe is formed.

θ = ...° [1]

(iii) On Fig. 5.2, sketch the variation of the intensity I with θ for values of θ from –15° to +15°. [3]

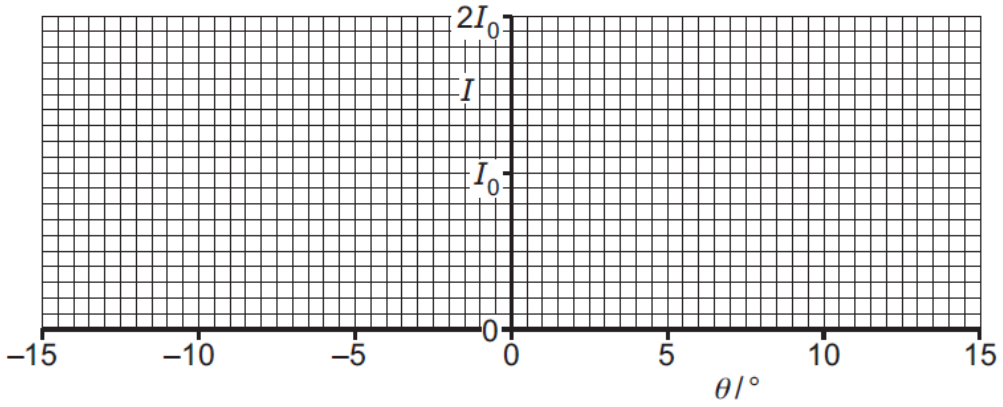


Fig. 5.2

(b) A polarising filter is placed in the path of the light beam that is incident on the diffraction grating in Fig. 5.1. The transmission axis of the filter is at 45° to the vertical.

State how the intensity at the peaks for the light on the screen compares with the answer in **(a)(iii).**

……..

……..

..... [1]

[Total: 6]

6 **(a)** Fig. 6.1 shows two resistors **X** and **Y** connected in series to a battery with an internal resistance. The voltmeter has an infinite resistance.

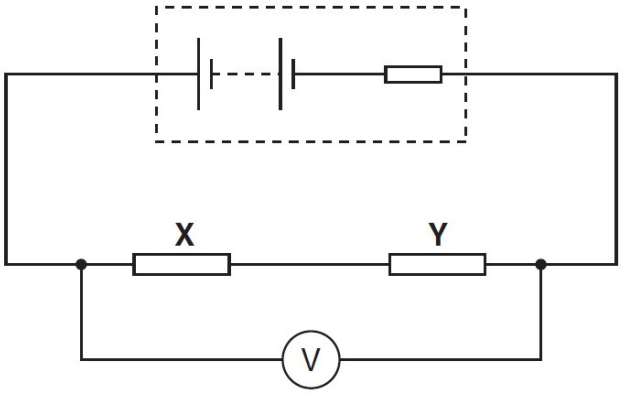


Fig. 6.1

(i) Explain why the voltmeter reading is not equal to the e.m.f. of the battery.

……

.

..... [1]

(ii) The resistance of resistor **X** is four times the resistance of resistor **Y**. The voltmeter reading is 6.0 V.

Calculate the potential difference across the resistor **Y**.

potential difference = …… V [2]

(b) Fig. 6.2 shows four capacitors, each of capacitance 50 μF, connected to a high-voltage supply, a resistor and a switch S.

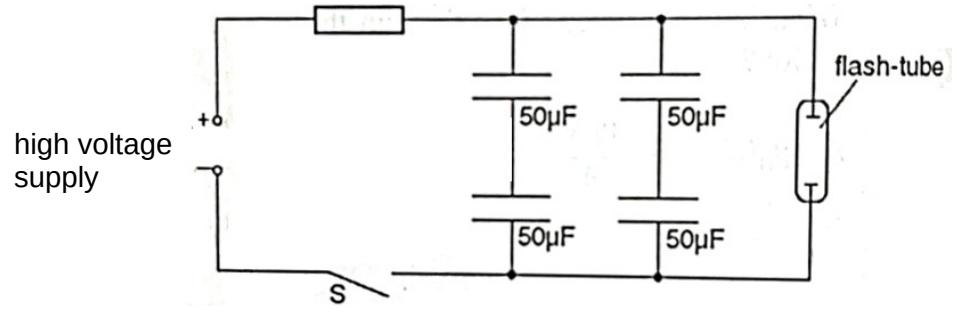


Fig. 6.2

A flash-tube is connected across the combination of capacitors. When S is closed, the capacitors charge up gradually and, when the potential difference across the flash-tube reaches 540 V, the capacitors discharge rapidly through the flash-tube.

A flash of light is emitted as the capacitors discharge 63% of their stored energy.

(i) Determine the total capacitance of the arrangement of capacitors.

capacitance = …….…….. μF [2]

(ii) Calculate

1. the energy dissipated in one flash of light,

energy = …….. J [2]

2. the potential difference across each capacitor immediately after the flash of light has occurred.

potential difference = …… V [2]

[Total: 9]

7 **(a)** A parallel beam of light of wavelength 4.5×10^{-7} m and intensity 700 W m^{-2} is incident normally on a surface.

(i) Calculate the number of photons incident per second on an area of 1.0×10^{-4} m^2 of the surface.

number = ……[2]

(ii) Calculate the momentum of a photon of the light.

momentum = ……N s [2]

(iii) Calculate the change in momentum of photons incident on 1.0×10^{-4} m^2 of the surface in one second. Assume that all photons are absorbed by the surface.

change in momentum = …… N s [2]

(b) The orbital electron in a hydrogen atom may be thought of as being trapped in an infinite square potential well such that the electron is confined to a region of width L, as shown in Fig. 7.1.

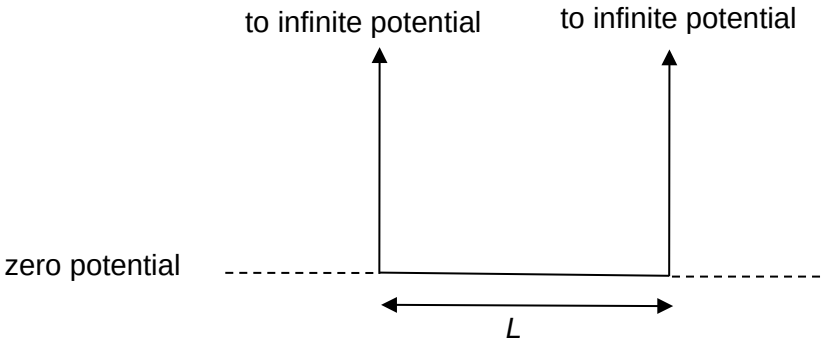


Fig. 7.1

The electron is represented by a standing wave that has wavefunction Ψ given by

$$\Psi = A \sin nkx$$

where x is the distance of the electron from the left-hand edge of the potential well, n is a non-zero integer and A and k are constants that depend on L.

(i) When $n = 1$, the electron is in its ground state.

Determine an expression for k in terms of L.

k = ……[2]

(ii) State what is represented by $|\Psi|^2$.

……..

..... [1]

(iii) Determine an expression for *A*. Show your working.

[2]

[Total: 11]

Section B

Answer **one** question in this section in the spaces provided.

8 **(a)** **(i)** Define simple harmonic motion.

……..

……..

..... [2]

(ii) A dock has a tidal entrance at which the water is 10 m deep at 12 noon, when the tide is at its lowest. The water is 30 m deep at its highest, which occurs next at 6.15 pm. A tanker, needing a depth of 15 m, requires entering the dock as soon as possible that afternoon.

Calculate the earliest time it could just clear the dock entrance. Show your working.

time ……[6]

(iii) State what you have assumed about the motion of the tide.

.....

[1]

(b) Fig. 8.1 shows a horizontal copper wire placed between the opposite poles of a permanent magnet. The wire is in a state of tension and is clamped at each end.

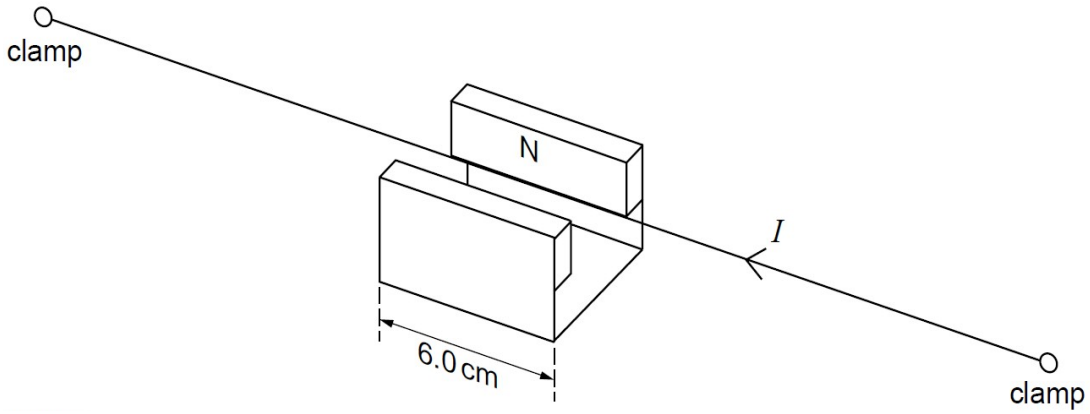


Fig. 8.1

(i) An alternating current I of constant amplitude and variable frequency is passed through the wire causing the wire to oscillate. Fig. 8.2 shows how the acceleration of the wire at the centre point between the poles varies with time when the frequency of the current is at the fundamental natural frequency of the wire.

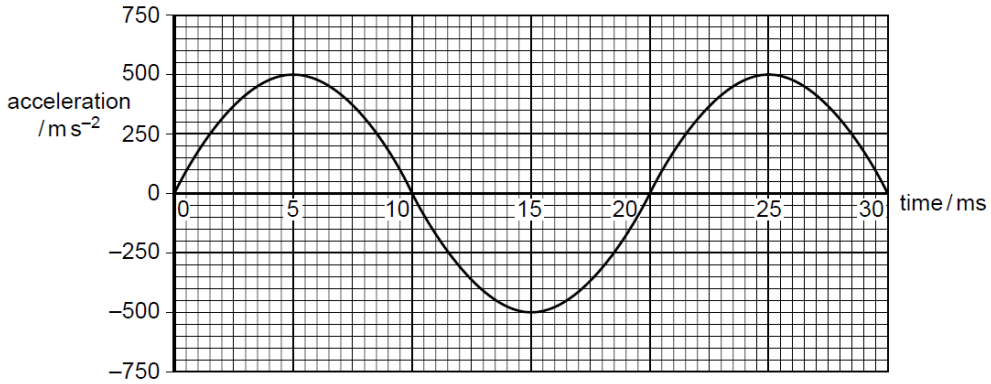


Fig. 8.2

1. Calculate the frequency of the alternating current.

frequency = …….. Hz [2]

2. Explain whether the maximum acceleration at all points on the wire between the poles is the same or not. A sketch may help your answer.

.....

.....

.....

..... [3]

(ii) The amplitude of vibration of the wire at the centre point between the poles varies with the frequency of the current as shown in Fig 8.3.

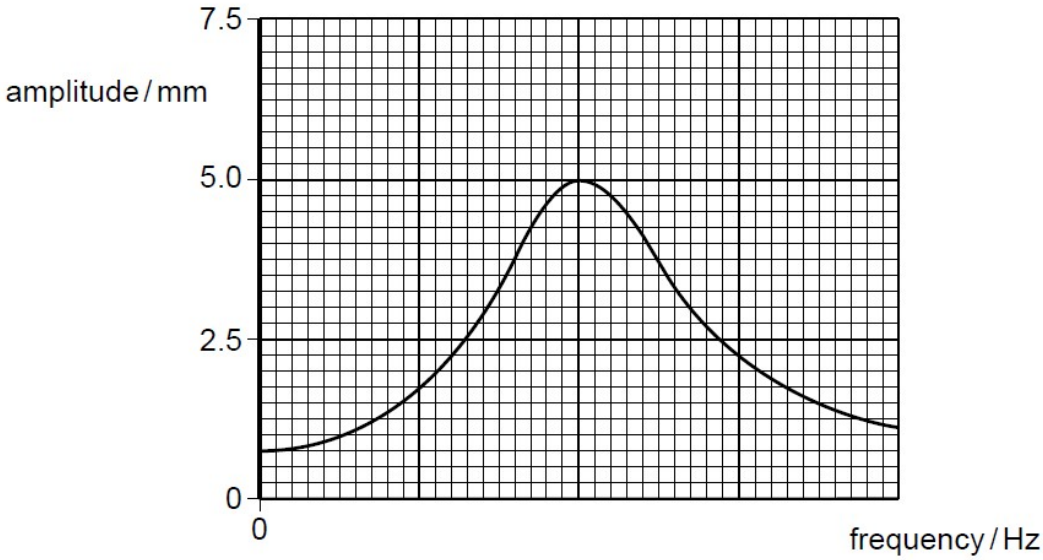


Fig. 8.3

1. State the term used to describe this effect.

……..…….....
[1]

2. Add the scale to the frequency axis on Fig 8.3. [1]

3. A student sticks some small pieces of paper to the wire to view the vibration of the wire more easily. This increases the damping of the vibration.

Explain what is meant by damping and on Fig. 8.3 above, sketch the graph you would expect to see for greater damping.

.....

…… [4]

[Total: 20]

9 **(a)** A radiation detector is placed close to a radioactive source.

The variation with time *t* of the measured count rate is shown in Fig. 9.1.

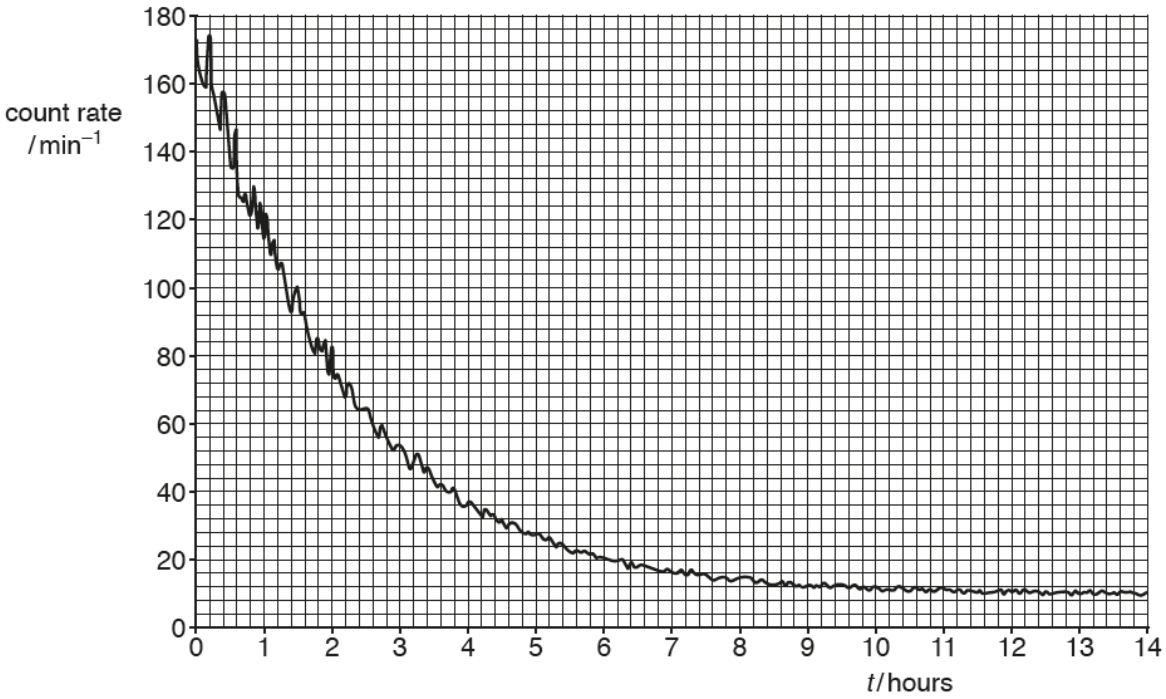


Fig. 9.1

The graph is used to determine the half-life of the radioactive source.

(i) Explain how to take into account the random nature of count rate in the graph in Fig. 9.1.

……..

..... [1]

(ii) Explain how to determine the background count rate from the graph in Fig. 9.1.

……..

..... [1]

(iii) Hence use Fig. 9.1 to determine the half-life of the radioactive source.

half-life = …… hours [3]

(iv) The readings in Fig. 9.1 were obtained at room temperature. A second sample of this radioactive source is heated to a temperature of 500 °C. The initial count rate at time t = 0 is the same as that in Fig. 9.1. The variation with time t of the measured count rate from the heated source is determined.

State and explain whether the half-life of the radioactive source changes when the temperature is increased to 500 °C.

……

.

……

.

..... [2]

(b) An unstable nucleus of nucleon number (mass number) A undergoes α-decay, as illustrated in Fig. 9.2.

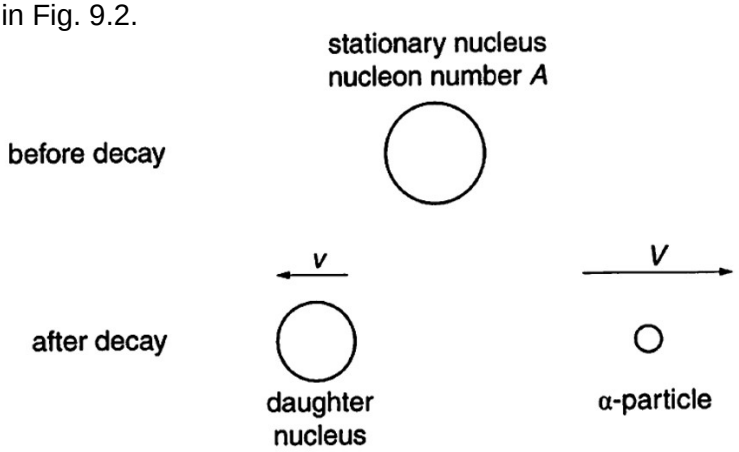


Fig. 9.2

The nucleus is stationary before the decay. After the decay, the initial speed of the α-particle is *V* and that of the daughter nucleus is *v*.

(i) State an equation, in terms of *A, v* and *V*, to represent conservation of linear momentum for this decay.

……

.

..... [1]

(ii) Show that the ratio

$$\frac{\text{initial kinetic energy of } \alpha\text{-particle}}{\text{initial kinetic energy of daughter nucleus}}$$

is equal to $(\frac{1}{4}\ A\ -\ 1)$.

[3]

(c) Data for the α-decay of bismuth-212 ($^{212}_{83}Bi$) to form thallium-208 ($^{208}_{81}Tl$) are given in Fig. 9.3.

nucleus	mass of nucleus / u
bismuth-212	211.9459
thallium-208	207.9374
helium-4	4.0015

Fig. 9.3

(i) Use the data of Fig. 9.3 to calculate, to two decimal places, the energy released during the decay.

energy = …… MeV [4]

(ii) Use your answer in **(c)(i)** to show that, based on the expression in **(b)(ii)**, the energy of the α-particle is 6.42 MeV.

[2]

(d) In practice, the α-particle is found to have an energy of 6.10 MeV, rather than the 6.42 MeV shown in **(c)(ii).**

Suggest

(i) an explanation for the difference in energy,

……..……..

..... [1]

(ii) why is it unlikely that the thallium nucleus and the α-particle move off in opposite directions.

……

.

……

.

……

.

..... [2]

[Total: 20]

End of Paper